EXPLORATORY DATA ANALYSIS (EDA) PROJECT

Iris Flower Dataset

This project explores the Iris dataset using statistical summaries, visualizations, and correlation analysis to identify patterns, trends, and important influencing factors.

1. Objective

The main objective is to understand the structure of the dataset, summarize its numerical features, visualize important relationships, and identify which measurements are strongly related to one another and to flower species.

2. Dataset Description

The dataset contains 150 flower observations from three species: setosa, versicolor, and virginica. Each observation has four numerical measurements: sepal length, sepal width, petal length, and petal width.

Feature	Description
sepal_length	Sepal length in centimeters
sepal_width	Sepal width in centimeters
petal_length	Petal length in centimeters
petal_width	Petal width in centimeters
species	Flower species

3. Statistical Summary

Feature	Mean	Std Dev	Min	Max
sepal_length	5.84	0.83	4.30	7.90
sepal_width	3.06	0.44	2.00	4.40
petal_length	3.76	1.77	1.00	6.90
petal_width	1.20	0.76	0.10	2.50

4. Data Visualizations

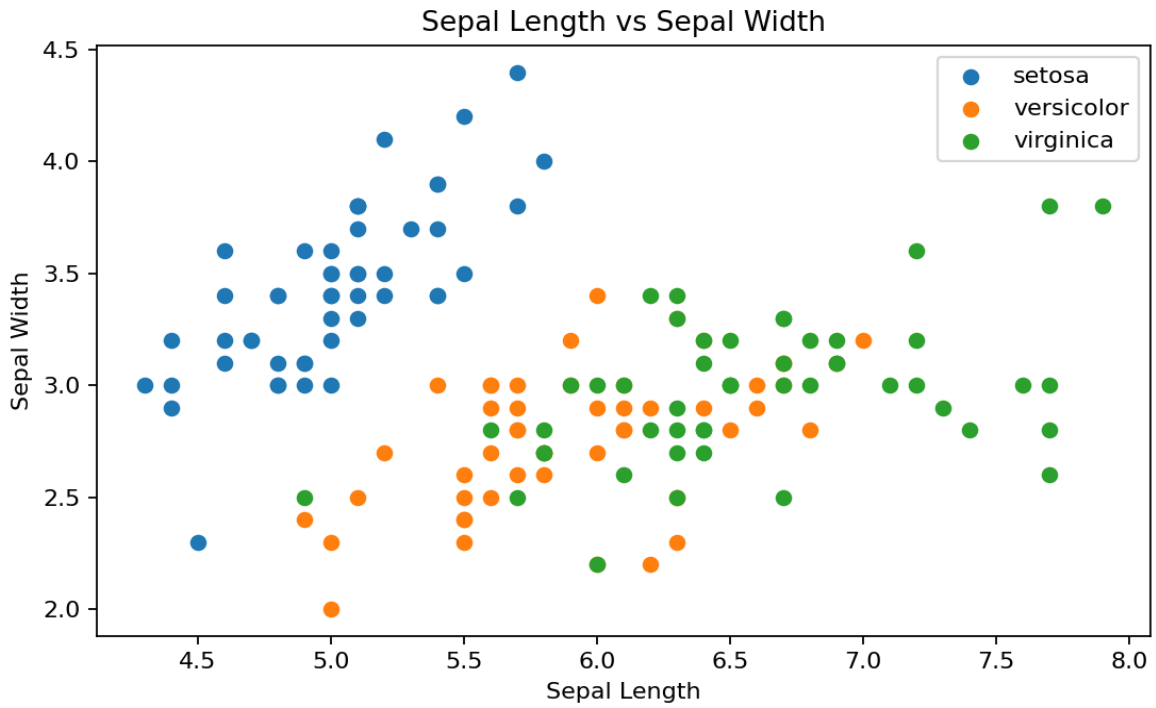


Figure 1: Relationship between sepal length and sepal width.

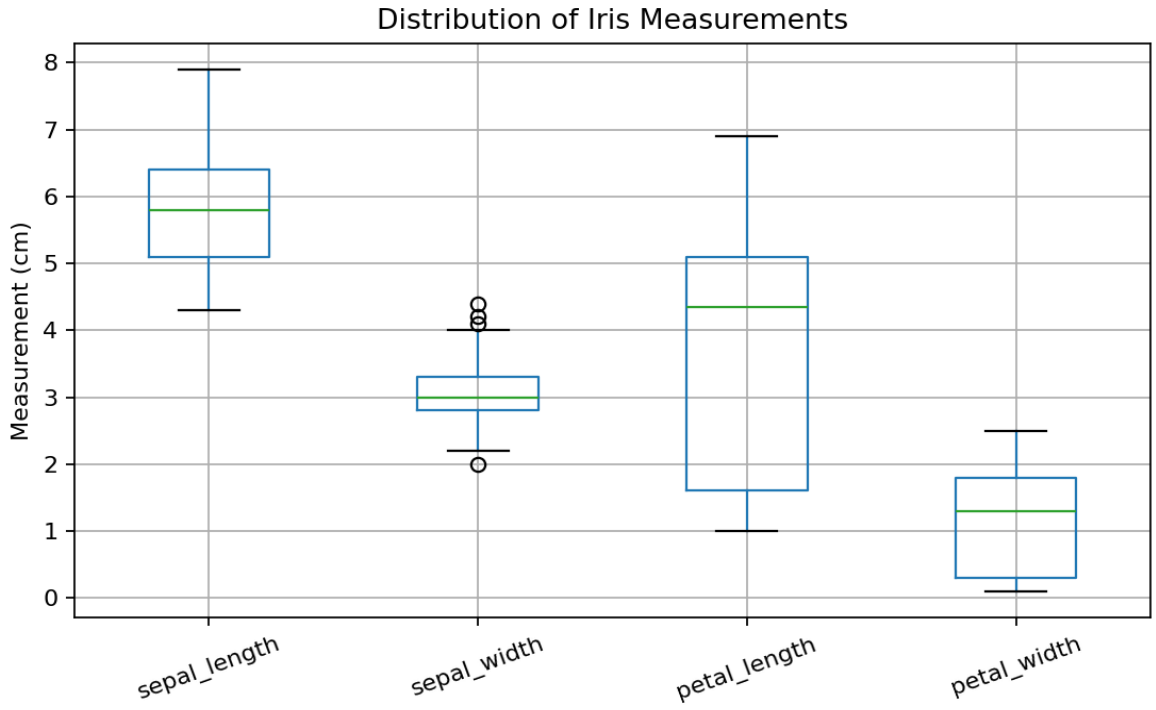


Figure 2: Distribution and spread of the four measurements.

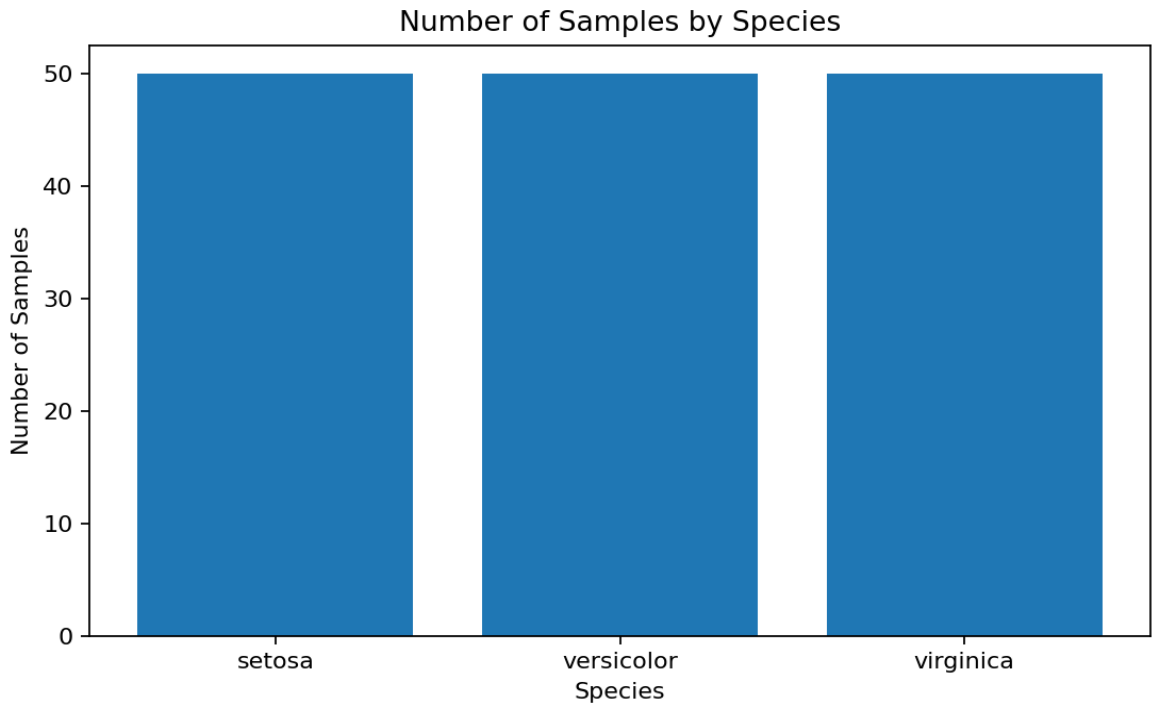
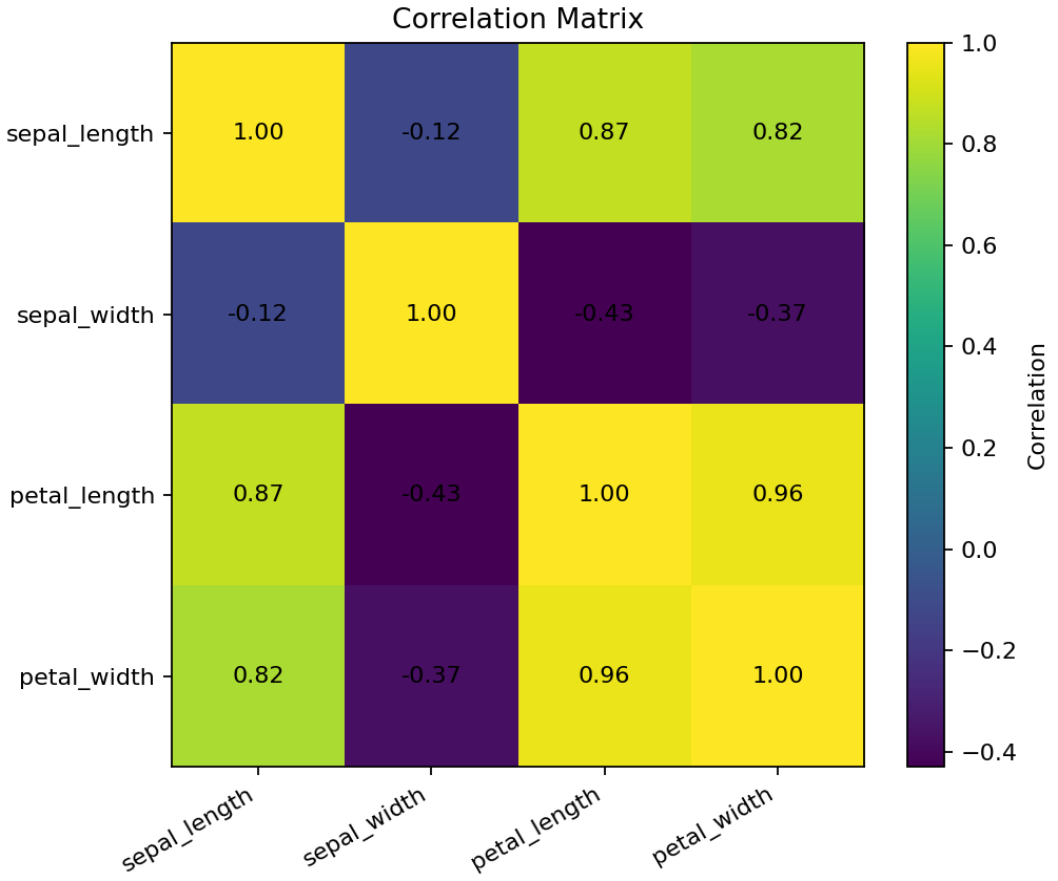


Figure 3: Number of observations for each species.

5. Correlation Analysis

	sepal_length	sepal_width	petal_length	petal_width
sepal_length	1.00	-0.12	0.87	0.82
sepal_width	-0.12	1.00	-0.43	-0.37
petal_length	0.87	-0.43	1.00	0.96
petal_width	0.82	-0.37	0.96	1.00



6. Key Findings

• Petal length and petal width show a strong positive relationship: flowers with longer petals generally also have wider petals.

• Sepal length also has a noticeable positive relationship with petal measurements.

• Sepal width has a weaker relationship with the other measurements compared with petal length and petal width.

• The three species have the same number of observations, so the dataset is balanced by species.

• The scatter plot shows that petal-based measurements provide clear separation between species, while sepal measurements overlap more.

• The box plot helps identify the range and spread of each measurement and provides a visual way to inspect possible unusual values.

7. Conclusion

The EDA shows that the Iris dataset contains clear relationships among flower measurements. Petal length and petal width are particularly informative features for distinguishing species. Statistical summaries provide the

numerical overview, while graphs make the relationships and differences easier to understand. Overall, the analysis demonstrates how EDA can convert raw data into useful patterns and insights before applying further data-science methods.

8. Tools Used

Python, Pandas, NumPy, Matplotlib, Scikit-learn, and ReportLab were used to load, analyze, visualize, and document the dataset.